

No Pump-Free Corner: The Residual Doors of Covariant Modified Inertia, Computed — and a Pre-Registered Sign-Flip Signature for Dynamically Unsettled Galaxies

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Abstract

The six-theorem chain on covariant modified inertia (MI) — Ostrogradsky, Cassini/slip, the sign/state clause, the frequency law, the gas clamp, and the finite-time/non-stationary closure (Zenodo 10.5281/zenodo.21016309, 21139029, 21148494, 21152331, 21175723) — named its own residual doors rather than declaring the territory sealed. This note computes all four. **(D1) The interacting-field window** (the named mind-changer of Theorem VI): $F(\omega) \geq 0$ is a theory-independent Gram fact; a new monotone-window lemma extends Bisognano–Wichmann detailed balance to arbitrary window lengths on uniform trajectories for any interacting Wightman QFT; and a Källén–Lehmann reduction — independently re-derived on a genuine two-particle composite with zero fitted constants (agreement 1e-4) — expresses the interacting vacuum’s windowed response on *any* trajectory as a positive mixture of free massive-slice responses, none of which invert (the sharp-kick margin asymptotes to zero from below; equality, never inversion). Driven interacting benchmarks do invert (11.4×) — and the free control inverts 407× under the same drive: agent-pumped Raman gain, the already-priced pump channel. Closed at 2nd order in detector coupling, the order at which the theorem’s object is defined; the honest residuals are 4-point effects and dS Bros–Moschella positivity. **(D2) The passive-anharmonic corner** — the last pump-free route: passivity *alone* (KMS not needed) forces $\delta m \geq 0$ in linear response at any anharmonicity, non-perturbatively in coupling for adiabatic worldlines (an exact term-by-term pair sum, re-derived independently to 0.58% with gap-sweep exponent -3.000000), with the degeneracy edge sealed. The adversarial verifier’s open-direction attack *partially succeeded* — instructively: at strong drive near the two-photon resonance ($gA=5$, $2\Omega \approx \text{gap}$) the reactive dressing genuinely goes negative from a passive start; the autopsy shows the drive de-passivizes the bath (the drive *is* the pump; Pusz–Woronowicz never violated), the window is frequency-locked, and band-mapping requires bath gaps $\lesssim 10^{-31}$ eV — the gas clamp. Amended closure: the passive softening channels are the sub-drive pole and its multiphoton/Floquet generalization, both drive-pumped, both clamped. **No pump-free MOND-sign channel exists.** **(D4) The flat-curve exceptional-point closure** of Theorem IV, done in full with Coriolis: the PT-broken band on flat curves is exactly $\mu \in (0.7579886, 0.8947874)$ — correcting our own published estimate 0.854, which sits *inside* the band — with a re-entrant real window (0.5, 0.758) the scalar shortcut missed; an exact fold law $\mu_{\text{fold}} = 1 - 1/(2p)$ shows sub-quadratic PU kernels ($p=1/2$) evade the EP entirely (deep-MOND PU orbits exist and are linearly stable — a genuine soft spot in the published Fourth Horn’s blanket claim), and those kernels die instead by profile inversion (required vs delivered $(1-\mu)$ slopes are sign-opposite;

×49–80 on realistic disks). The horn stays dead; one of its bricks is relabeled in print. **(D3) The one door that opens:** fusing the Theorem-VI transient with the framework’s memory-kernel σ -spread yields a pre-registered, dated prediction table for dynamically unsettled systems — Crater II +13.6–26.5%, Antlia II +3.4–6.7%, settled controls <2%, and the clean MI-unique signature: **first-infall systems run 11–21% cold, flipping to hot after pericentre** — a relational sign flip that modified gravity (exactly zero by theorem) and tidal heating (no coherent deficit-then-excess) cannot mimic. Decisive at $3\sigma \approx 2029\text{--}2031$ (Gaia DR4 + WEAVE/4MOST dwarfs; CHANCES cluster infall populations). All lanes carry both coefficient footings ($a_0 = 9.36 \times 10^{-11}$ canonical vs 1.13×10^{-10}); no verdict flips. Verifier corrections are printed beside the results they amend, including two corrections to our own prior publications.

1. Four named doors, four computations

A no-go program earns trust by naming its own edges and then walking to them. Theorem VI left four: one theoretical mind-changer (interacting fields), one pump-free corner (passive anharmonic baths), one bookkeeping debt (the flat-curve EP), and one phenomenological opening (unsettled populations). Method as throughout this program: independent compute lanes, each attacked by an adversarial verifier instructed to refute in *both* directions; every load-bearing number backed by a committed script (numpy/sympy, no network, exit 0); the recurring footing fork ($a_0 = cH_\Lambda/Z = 9.36 \times 10^{-11} \text{ m/s}^2$, ρ_{DE} , vs 1.13×10^{-10} , ρ_{total}/cH_0) carried everywhere.

2. D1 — the interacting-field window closes at second order

Theorem VI’s positivity locks ($F(\omega) \geq 0$; $F(-\omega) - F(\omega) = (\omega/2\pi)\int \chi^2$) were proven for free fields pulled back to worldlines. Could an *interacting* vacuum, probed through a window on a non-uniform trajectory, invert?

Three closures and one honest benchmark:

1. **Positivity is universal.** $F(\omega)$ is the squared norm of a smeared field acting on the state — a two-line Gram argument valid for any state of any QFT. Confirmed to machine precision in a nonintegrable transverse-field Ising chain (static and moving probes; vacuum, excited, random, thermal states).
2. **Uniform trajectories, any interacting Wightman QFT.** Bisognano–Wichmann gives wedge-KMS regardless of interaction. A **monotone-window lemma** (new, proved, verified) then shows the detailed-balance inequality $F(\omega) \leq F(-\omega)$ survives *any* window length for Gaussian-class windows: max deviation $\sim 2.7 \times 10^{-2}$ over the KMS-spectrum $\times \beta \times$ window-length scan, with the β -exponential recovered to five digits in the long-window limit. The lone counterfeit — boxcar windows can fake ratio ~ 23 by parking a sinc zero on a narrow line — vanishes for Gaussian windows ($\sim 10^{-118}$) and for all continua: a windowing artifact, not physics.
3. **Non-uniform trajectories.** Pusz–Woronowicz honestly does *not* close these (the external agent moving the detector supplies work — the Unruh effect itself excites detectors). So: compute. The load-bearing step is a **Källén–Lehmann reduction**: at 2nd order in detector coupling — the order at which Theorem VI’s F is defined — the interacting vacuum’s windowed response on *any* trajectory is a positive mixture of free massive-slice responses. The verifier

re-derived this from scratch on a genuinely composite probe (ϕ^2 ; a two-particle continuum, KL density $\rho = 1/(\pi\mu^2\beta)$ obtained via the center-of-mass Jacobian) and confirmed the mixture against the direct position-space response on a non-uniform burst to $1-4\times 10^{-4}$ **with zero fitted constants**. The free massive slices were then scanned over 28 burst/oscillation/kick configurations, and adversarially over 7 more beyond the grid (light masses, sharp fast bursts, chirps, off-center windows, resonance-tuned oscillations): **no inversion anywhere**. The sharpest kicks drive the no-inversion margin monotonically toward zero *from below* ($-4.6\times 10^{-2} \rightarrow -1.9\times 10^{-2}$): the sudden limit approaches equality and never crosses — in 3+1D the $k^2 dk$ phase space never blocks the emission resonances that a discrete lattice can.

4. **The driven benchmark keeps the bookkeeping honest.** A nonperturbatively driven interacting lattice (Kim–Huse class, $L=9$) *does* invert — $F(\omega)/F(-\omega) = 11.4$ at the drive-Stokes sideband $\omega^* = 2\Omega_{\text{drive}} - \Delta E$. But the free-fermion control under the identical drive inverts **407×**. Inversion is not interacting-specific; it is agent-pumped Raman gain — precisely the pump channel the chain already prices at $\times 2.9\times 10^{10}$ shortfall.

Verdict: CLOSED at 2nd order. The genuinely open remainders, named and not claimed: 4-point-function (4th-order) effects, where the KL reduction is silent; and Bros–Moschella dS positivity for light/complementary-series interacting fields, unresolved in the mathematical literature.

3. D2 — passivity alone forces the anti-MOND sign: the pump requirement is fully general

The last pump-free hope: Theorem III’s $\delta m \geq 0$ was proven for harmonic/two-level passive baths; anharmonic baths escape *state-blindness* — could a passive-but-anharmonic medium carry $\delta m < 0$ with no power source?

No — and now at theorem grade:

1. **Linear response, any anharmonicity, any stationary passive state.** In the joint ρ/H eigenbasis every Bohr-line weight is $(p_n - p_m)|B_{nm}|^2 \geq 0$ under the passivity ordering, so $\omega \cdot \text{Im}\chi(\omega) \geq 0$, and the worldline dressing $\delta m = (2/\pi)g^2 \int \text{Im}\chi/\omega^3 \geq 0$. **KMS is not needed** — passive-but-not-KMS states are covered directly. (Minimum Bohr-line weight over 2000 random anharmonic passive ensembles: $+1.2\times 10^{-8}$; over the passive polytope’s extreme points: 0 exactly, term-by-term.)
2. **Non-perturbative in coupling** (adiabatic worldlines): the $O(v^2)$ mass coefficient is the exact pair sum $2g^2 \sum (p_n - p_m) |\langle m | \partial H / \partial x | n \rangle|^2 / \omega^3$ over the *full displaced* spectrum — term-by-term ≥ 0 for states passive with respect to the instantaneous Hamiltonian, a premise the verifier tested directly (exact slow sweep: zero ordering violations along the trajectory; finite-speed reordering is Landau-Zener work, i.e. drive-paid). Exact-diagonalized quartic scans across coupling, displacement, and passive-state families: global minimum $+3.2\times 10^{-4}$, while the most adversarial *non*-passive state reaches -0.0038 — passivity, not model structure, is what binds. The lane caught and killed its own fake counterexample (a Fock-truncation artifact printing $\delta m = -213$; converged truncation $\rightarrow$ all-passive positive). The verifier re-derived the pair-

sum law on a different anharmonic model, truncation, and integrator: direct dynamical mass measurement agrees to 0.58%, gap-sweep log-log slope -3.000000 , harmonic anchor to 6×10^{-17} .

3. **The degeneracy edge** — passivity permits arbitrary coherences inside degenerate subspaces, the sharpest crack candidate: such coherences commute with the block Hamiltonian and contribute constant geometric forces with *zero* inertia (force-variation 9×10^{-13}); 300 coherent passive states all harden (min $+0.52$); near-degenerate passive pairs harden as $+1/\omega_0^3$.

The verifier’s catch — an open-direction attack that partially succeeded, and what it teaches. Beyond the lane’s scan range (drive amplitude $gA=5$ at the two-photon resonance $2\Omega \approx \text{gap}$), the reactive dressing **genuinely goes negative from a passive start** ($\delta m = -0.0169$; -0.007 from thermal $T=2$). The lane’s “never crosses zero” was a scan-range artifact — corrected in print here. The autopsy is the point: (i) the effect is frequency-locked to a narrow window ($\Omega/\text{gap} \in [0.40, 0.56]$; positive again by 0.60) — a *frequency* law, never $\mu(a/a_0)$; (ii) the bath *absorbs* energy throughout ($+5.07$ per protocol; Pusz–Woronowicz never violated to $gA=5$); (iii) the drive de-passivizes the bath before softening appears (6 population-ordering violations vs 0 initially) — **the drive is the pump**; (iv) mapped to the galactic band it requires bath gaps $\lesssim 10^{-31}$ eV, where the gas clamp (Theorem V) applies exactly as it does to the sub-drive pole.

Amended closure (the citable sentence): passivity alone forces $\delta m \geq 0$; the only passive-start softening channels are the sub-drive pole *and its multiphoton/Floquet-dressed generalization* — both frequency-locked, both drive-pumped, both clamped by Theorems IV/V. **There is no pump-free MOND-sign channel.** Footing-independent (spreads printed; 20.8% on a_0 , no flips).

4. D4 — the flat-curve EP, exactly: a published number corrected, a soft spot found, the horn dead by a different brick

Theorem IV (Zenodo 10.5281/zenodo.21148494) proved its exceptional-point cap $\mu \geq 1/2$ harmonic-exactly, and carried a crude flat-curve estimate ~ 0.854 “pending the Coriolis linearization.” Done in full — the radial–azimuthal PU linearization around a circular orbit in the self-consistent flat-curve potential ($\kappa = \sqrt{2}\Omega$), degree-8 characteristic polynomial, cross-checked by companion-matrix eigenvalues *and* nonlinear orbit integration (measured runaway 0.296Ω vs 0.300 predicted, 1.5%):

- **Exact flat-curve band:** PT breaks for $\mu \in (0.7579886, 0.8947874)$ — the roots of $664\mu^4 - 3036\mu^3 + 5086\mu^2 - 3681\mu + 971$. The published 0.854 sits *inside* the band: wrong in both directions. There is a re-entrant PT-real window $(0.5, 0.758)$ the scalar treatment missed; below $\mu = 1/2$ the spectrum is doubly unstable (real saddle + complex quartet), so deep-MOND orbit non-existence is *confirmed for the quadratic proxy*. Family map across rotation-curve shapes $\kappa/\Omega \in [1, 2]$: the upper band edge is nearly universal (0.892 – 0.9); the harmonic case is the unique never-breaking endpoint — explaining retroactively why the harmonic-exact cap missed the band entirely.
- **The fold law and the genuine soft spot.** For general PU kernels $k(y)$, the fold obeys the exact factorization $\mu_{\text{fold}} = 1 - 1/(2p)$. Sub-quadratic kernels — including the framework-tail case $p = 1/2$ — are **PT-real at all $\mu < 0.5486$** : deep-MOND PU orbits *exist and are linearly stable* there. The Fourth Horn’s blanket “deep MOND fails by orbit non-existence” was quadratic-proxy-

specific — a correction to our own published theorem’s *mechanism*, found by this program’s both-ways rule and printed here. (The verifier independently confirmed the fold law nonlinearly at $p = 3/4$ — fold exactly $\mu = 1/3$ — and confirmed the soft spot is not a degeneracy artifact.)

- **Why the horn stays dead:** those evading kernels force $d \ln(1-\mu)/d \ln R = -2p$ along a flat curve, while the framework’s own v requires $\approx +1$ — sign-opposite slopes, a mismatch of $\times 49$ –**80 on realistic 2–20 kpc disks** ($p=1/2$; $\times 493$ – 799 for $p=1$; verifier-corrected from the $\times 100$ asymptotic bound; $\sim 10\%$ footing spread, no flips). Profile inversion does the work the EP was credited with.
- **Scope:** the PU horn only. The framework’s own nonlocal-kernel MI and its phenomenology are untouched by this lane.

5. D3 — the door that opens: a pre-registered sign-flip signature for unsettled galaxies

Theorem VI quantified a residual: one-shot transients contribute at the percent level per decade of orbits — invisible in settled disks (the RAR), potentially distinctive where relaxation is recent. Fused here with the framework’s banked memory-kernel physics (Lorentzian kernel, $\tau = 0.45$ Gyr, $\theta_0 \in [\sqrt{2}, 2]$) into one amplitude-vs-settledness observable, computed on the framework’s own objects, with the corpus’s own orbit machinery, and corrected by adversarial verification (four corrections folded; the two-channel double-count and a factor-2 print bug removed — the table below is the defensible, memory-channel-only version).

The pre-registered table (relational σ signal vs settled expectation; both footings, no flips):

System / class	Predicted signal	Phase context
Crater II	+13.6 to +26.5%	post-pericentre ($t_{\text{peri}} = 0.77$ Gyr, phase 0.27)
Antlia II	+3.4 to +6.7%	near apocentre (phase 0.96)
Sagittarius	+23 to +27%	tidally dirty — excluded as a clean carrier
First-infall dwarfs / cluster members	–11 to –21% (deficit)	before first pericentre — the sign flip
Leo I	≈ 0	density-gate null control
Settled controls (Fornax/ Sculptor class)	< 2%	
Cluster relational spread (re- cent-infall vs virialized)	+6 to +13%	recovers the banked MI band

The clean MI-unique signature is the sign flip, not the decay clock: the verifier showed the $\tau \approx 0.45$ Gyr exponential decay is degenerate with post-shock tidal revirialization on diffuse carriers ($t_{\text{cross}} = 0.86 \cdot \tau_{\text{mem}}$ for Crater II — structurally forced for $y \sim 1$ carriers). What survives

is stronger: **a coherent σ deficit on first infall that flips to an excess after pericentre**. Tidal heating cannot produce a coherent pre-pericentre deficit; modified gravity and instantaneous- μ MI predict exactly zero relational effect (Milgrom’s MG-virial universality); Λ CDM+tides produce excess-only with hysteresis. The Luo window tail itself is restated honestly as a one-sided $\approx$ envelope (0.4–4.0% for $n_{\text{orb}} = 10 \rightarrow 1$) — no standalone test, since a null cannot falsify an envelope.

Decisiveness: matched-pericentre, matched-density phase comparisons. MW-dwarf arm: ~ 9 carriers/arm at 10% signal and 7% σ -error $\rightarrow 3\sigma \approx$ **2029–2030** (Gaia DR4 orbital phases, Dec 2026, + WEAVE/4MOST velocity dispersions). Cluster arm: 6% signal vs 2% bin error $\rightarrow 3\sigma \approx$ **2028–2031** (CHANCES recent-infall vs virialized populations). Kill conditions, stated now: DR4 revising Crater II/Antlia II orbital phases beyond ~ 2 Gyr post-pericentre kills the carrier liveness; a computed kernel shrinking the θ band below 5% kills the amplitude; tidal simulations producing a coherent *pre-pericentre deficit* with a pericentre flip would destroy the signature’s uniqueness.

6. Standing

- **The pump requirement is now fully general** (D2): no passive medium of any structure, probed any way — stationary, transient, windowed, driven-from-passive — carries the MOND sign without a pump, and every named pump is priced and clamped. Combined with D1, the sign wall now runs: free fields (state-blind, all times) $\rightarrow$ structured passive baths (theorem, all anharmonicities) $\rightarrow$ interacting vacua (positive-mixture reduction, 2nd order) $\rightarrow$ pumped channels (priced $\times 2.9 \times 10^{10}$, gas-clamped).
- **Theorem IV’s flat-curve bookkeeping is exact** (D4), its published estimate corrected, its conclusion re-derived through profile inversion — the kill survives its own mechanism’s correction.
- **The framework gains one new dated, falsifiable, MI-unique observable** (D3): the first-infall sign flip, decisive ~ 2029 – 2031 — joining $\bar{s}^{\text{TX}}$ (10.5281/zenodo.21137568), Gaia DR4 wide binaries, and DESI DR3 on the live ledger.
- **Doors that remain open, named:** 4-point/4th-order interacting effects; dS Bros–Moschella positivity; the D3 table awaiting data; the empirical fronts. Per this program’s standing rule, there is no “theory closed” — there is a map with computed edges.

7. Verification and scope

Every lane was adversarially verified by an independent agent instructed to attack in both directions; two verifier runs died on infrastructure and were re-run (their targets’ scripts all execute in < 10 s, exit 0). Corrections are printed beside the results they amend: D1’s number-provenance slip and grid-limitation sharpened (margin $\rightarrow 0^-$, never crossing); D2’s “never crosses 0” corrected by a partially-successful attack whose autopsy *strengthened* the closure (the Floquet channel named and clamped); D3’s factor-2 bug, envelope double-count, and decay-clock degeneracy removed (the printed table is the corrected one); D4’s asymptotic $\times 100$ margin brought down to the realistic $\times 49$ – 80 . Two of these amend our *own* prior publications (the Fourth Horn’s flat-curve estimate and its orbit-non-existence mechanism) — reported here with the same weight as the wins.

Scope: this note computes the named residual doors of the six-theorem chain. It does not claim “modified inertia is impossible” beyond the mapped territory; it does not derive a_0 ’s value or the MOND sign (the sign requires a pump; no candidate exists); and it does not touch the $a_0 = c^2\sqrt{(\Lambda/32\pi)}$ reframing or its live empirical tests. The author retracted all earlier theory-of-every-thing claims (2026-06-23); this note claims computed edges and one new dated prediction, not a completion.

Script index

Door	Scripts (real_research/reviews/ residual_doors_2026_07/, all exit 0)
D1	D1_1_general_positivity.py, D1_2_kms_window_pw_theorems.py, D1_3_kl_reduction_burst_scan.py, D1_4_interacting_lattice_benchmark.py, verify_D1_interacting_window_rederive.p
D2	D2_1_theorem_passivity_dispersion.py, D2_2_quartic_nonperturbative.py, D2_3_degeneracy_edge.py, verify_D2_passive_anharmonic_rederive.py, verify_D2_rederive_final.py
D3	D3_amplitude_vs_settledness.py, D3_named_systems_forecast.py, D3_discrimination_detectability.py, verify_D3_unsettled_observable_rederive.p
D4	D4_flat_ep_cap.py, D4_curve_family.py, D4_tail_kernel_kill_margin.py, verify_D4_flatcurve_EP_rederive.py

Ledger: real_research/RESIDUAL_DOORS_VERDICT_2026-07.md.

References (chain)

- Zimmerman, C.P., *Scale Without Law*, 10.5281/zenodo.21016309
- Zimmerman, C.P., *The Sign Premise Is a State Clause*, 10.5281/zenodo.21139029 (Theorem III)
- Zimmerman, C.P., *The Fourth Horn*, 10.5281/zenodo.21148494 (Theorem IV — flat-curve estimate corrected herein)
- Zimmerman, C.P., *The Kernel That Builds Its Own Laser*, 10.5281/zenodo.21152331 (Theorem V)
- Zimmerman, C.P., *Scale Yes, Shape Yes, Sign No*, 10.5281/zenodo.21175723 (Theorem VI — residual doors named)
- Zimmerman, C.P., *Orbital History Writes the Cosmic Clock on Dwarf Galaxies* (v3), 10.5281/zenodo.20969585 (the memory kernel)
- Luo, M.J., arXiv:2602.14515 (2026)
- Milgrom, M., arXiv:1211.4899, 1311.2579, 2208.07073 (MG-virial universality; the relational discriminator’s parent machinery)
- Bisognano, J.J. & Wichmann, E.H., J. Math. Phys. 16, 985 (1975)

- Pusz, W. & Woronowicz, S.L., Commun. Math. Phys. 58, 273 (1978)
- Bros, J. & Moschella, U., Rev. Math. Phys. 8, 327 (1996)